

Ratio test - 5.6

$$\sum_{n=0}^{\infty} ar^n$$

Recall: a geometric series $a + ar + ar^2 + ar^3 + \dots$ converges if and only if $|r| < 1$

Idea: if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = |r|$, then

$$a_{n+1} \approx r a_n$$

Ratio test: Let $\sum_{n=1}^{\infty} a_n$ be a series with nonzero terms, and let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

- i. If $0 \leq L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.
- ii. If $L > 1$ or $L = \infty$, then $\sum_{n=1}^{\infty} a_n$ diverges.
- iii. If $L = 1$ exactly, the test is inconclusive.

Example: Analyze the convergence of $\sum_{n=1}^{\infty} \frac{n^2+1}{2^n}$.

Ratio test:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^2+1}{2^{n+1}}}{\frac{n^2+1}{2^n}} \right| \\&= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2+1}{2^{n+1}} \cdot \frac{2^n}{n^2+1} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^2+1}{2(n^2+1)} \right| \\&= \lim_{n \rightarrow \infty} \left| \frac{n^2+2n+2}{2n^2+2} \right| = \frac{1}{2}\end{aligned}$$

This is less than 1, so this series

$\sum_{n=1}^{\infty} \frac{n^2+1}{2^n}$ converges absolutely by the
ratio test.

Example: ATCO $\sum_{n=1}^{\infty} \frac{1}{n!}$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n!}{(n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| = 0$$

$$\left(\frac{6!}{7!} = \frac{\cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}}{7 \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}} = \frac{1}{7}, \quad (n+1)! = (n+1) \cdot n! \right)$$

And 0 is less than 1, so $\sum_{n=1}^{\infty} \frac{1}{n!}$

converges absolutely by the ratio test.

Example: ATCO $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)n^n}{(n+1)^{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n^n}{(n+1)^n} \right| = \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1} \right)^n \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n}{n+1} \right)^{-n}} = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+1}{n} \right)^n}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^n} = \frac{1}{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n} = \frac{1}{e} < 1$$

and therefore $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converges absolutely by the ratio test.